Estimating Coefficients with Contrast-Coded Predictors.

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Using Contrast Codes in Practice

Student	Group	SAT
1	Course	580
2	Course	560
3	Course	660
4	Course	620
5	Course	600
6	Course	580
7	Course	590
8	Course	640
9	Course	620
10	Course	600
11	No Course	580
12	No Course	530
13	No Course	590
14	No Course	550
15	No Course	610
16	No Course	590
17	No Course	600
18	No Course	530
19	No Course	590
20	No Course	600

- If we want to test the efficacy of an SAT prep course, we would have a model comparison:
  - A: $y_i = \beta_0 + \beta_1(X_{1i}) + \epsilon_i$
  - C: $y_i = \beta_0 + 0(X_{1i}) + \epsilon_i$
- Note that in Model C we are hypothesizing there is no effect of an SAT prep course.
  - In Model A, we are allowing for a non-zero effect.

Error and 'Fitting' Model C

Student	Group	SAT	X_1	$\widehat{y_{ci}}$	e_{ci}	e_{ci}^2
1	Course	580	1	591	-11	121
2	Course	560	1	591	-31	961
3	Course	660	1	591	69	4761
4	Course	620	1	591	29	841
5	Course	600	1	591	9	81
6	Course	580	1	591	-11	121
7	Course	590	1	591	-1	1
8	Course	640	1	591	49	2401
9	Course	620	1	591	29	841
10	Course	600	1	591	9	81
11	No Course	580	-1	591	-11	121
12	No Course	530	-1	591	-61	3721
13	No Course	590	-1	591	-1	1
14	No Course	550	-1	591	-41	1681
15	No Course	610	-1	591	19	361
16	No Course	590	-1	591	-1	1
17	No Course	600	-1	591	9	81
18	No Course	530	-1	591	-61	3721
19	No Course	590	-1	591	-1	1
20	No Course	600	-1	591	9	81

$$\text{C: } y_i = \beta_0 + \epsilon_i$$

$$b_0 = \bar{y} = \frac{\sum y_i}{n}$$

$$\text{C: } y_i = 591 + e_i$$

Error and 'Fitting' Model A

Student	Group	SAT	X_1	$\widehat{y_{ai}}$	e_{ai}	e_{ai}^2
1	Course	580	1	605	-25	625
2	Course	560	1	605	-45	2025
3	Course	660	1	605	55	3025
4	Course	620	1	605	15	225
5	Course	600	1	605	-5	25
6	Course	580	1	605	-25	625
7	Course	590	1	605	-15	225
8	Course	640	1	605	35	1225
9	Course	620	1	605	15	225
10	Course	600	1	605	-5	25
11	No Course	580	-1	577	3	9
12	No Course	530	-1	577	-47	2209
13	No Course	590	-1	577	13	169
14	No Course	550	-1	577	-27	729
15	No Course	610	-1	577	33	1089
16	No Course	590	-1	577	13	169
17	No Course	600	-1	577	23	529
18	No Course	530	-1	577	-47	2209
19	No Course	590	-1	577	13	169
20	No Course	600	-1	577	23	529

$$\text{A: } y_i = \beta_0 + \beta_1(X_1)\epsilon_i$$

$$b_1 = \frac{\sum \lambda_k \bar{y}_k}{\sum \lambda_k^2}$$

$$b_1 = \frac{(-1)577 + (1)605}{(-1)^2 + (1)^2} = \frac{28}{2} = 14$$

$$\text{A: } y_i = 591 + 14(X_1) + e_i$$

Error and 'Fitting' Model A

Student	Group	SAT	X_1	$\widehat{y_{ai}}$	e_{ai}	e_{ai}^2
1	Course	580	1	605	-25	625
2	Course	560	1	605	-45	2025
3	Course	660	1	605	55	3025
4	Course	620	1	605	15	225
5	Course	600	1	605	-5	25
6	Course	580	1	605	-25	625
7	Course	590	1	605	-15	225
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18	No Course	530	-1	577	-47	2209
19	No Course	590	-1	577	13	169
20	No Course	600	-1	577	23	529

$$\text{A: } y_i = \beta_0 + \beta_1(X_1)\epsilon_i$$

$$b_0 = \frac{\sum \bar{y}_k}{m}$$

$$b_1 = \frac{\sum \lambda_k \bar{y}_k}{\sum \lambda_k^2}$$

$$b_0 = \frac{605 + 577}{2} = 591$$

$$\text{A: } y_i = 591 + 14(X_1) + e_i$$

Least Squares Estimates for Categorical Predictors

- In practice, we get statistical programs to obtain these parameters estimates for us.

- However, it is important to understand how these parameter estimates are obtained and how we should interpret them based on the coding of our variables.
  - Treatment/ ‘dummy’-codes are certainly valid, but contrast-codes offer a number of computational and interpretational advantages.

A t-test versus “the t-test”?

- Often, especially in undergraduate courses, ‘ the t-test’ might be taught as a specific test *only* for comparing two group-means.
  - Either implicitly or explicitly, students learn that t-tests are some thing different and distinct from regression.

- In the general linear modeling framework, however, we have shown that we can conduct a t-test in any situation in which we have a single degree of freedom in the numerator (i.e., PA – PC = 1).
  - We can conduct a t-test for an intercept,
  - for the slope of a continuous variable,
  - for the slope of a categorical variable (i.e., difference between group-means).